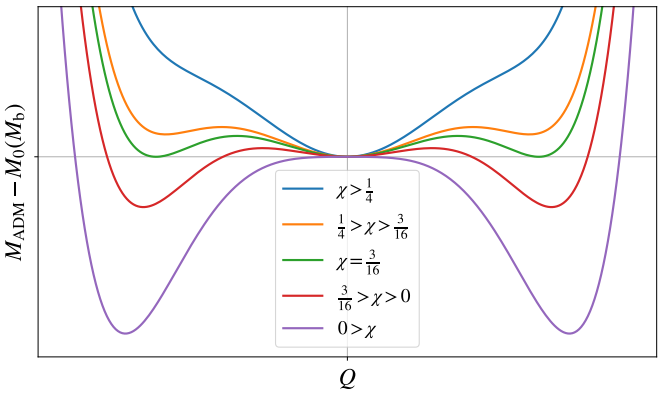

FIG. 4. The Landau ansatz in Eq. (3) in the case $b < 0$, plotted for different values of the parameter $\chi \equiv ac/b^2$. This is the case of a first-order phase transition.

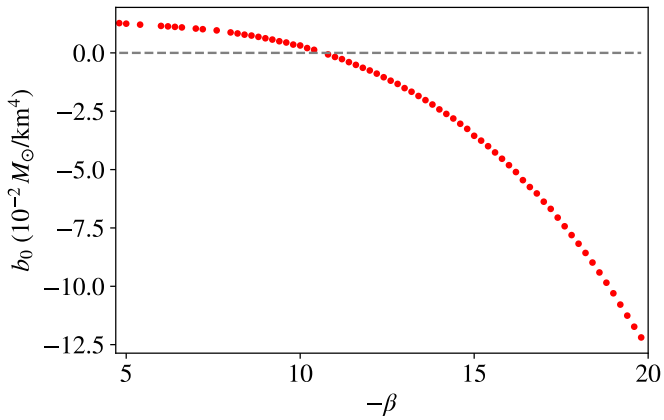

FIG. 5. $b_0 = b(M_{\rm crit})$ as a function of β for $m_\phi = 0$. The change from first- to second-order scalarization is near $\beta = -10.6$, that is, $b_0(\beta = -10.6) = 0$.

tinuously: physically relevant equilibria with arbitrarily small Q are possible. Recall that both signs of Q correspond to the same $M_{\rm b}(\tilde{\rho}_{\rm c})$ curve in Fig. 1.

This is a textbook example of a *second-order phase transition,* hence our nomenclature. In particular, $a(M_{\rm b})$ changes sign at a critical baryon mass where stable scalarized solutions appear, which is the $M_{\rm crit}$ from the previous section. This phenomenological explanation goes back to the earliest works on the topic [10, 35].

First-order scalarization occurs when $b(M_{\rm b})<0$, for which $c(M_{\rm b})>0$ is also relevant. The behavior of $M_{\rm ADM}(Q)$ changes drastically depending on the parameter $\chi \equiv ac/b^2$ that decreases with increasing $M_{\rm b}$, as seen in Fig. 4. There is a single stable equilibrium at $Q = 0$ when χ is positive and high, which becomes unstable for negative χ for which two symmetric stable equilibria exist. In between these phases, we have three local minima corresponding to stable or metastable states, as well as two symmetric local maxima that correspond to unstable equilibria. Quantitative details can be found in the "End Matter."

The phase transition picture in Fig. 4 elucidates many points in our numerical results which previously had no clear reason. For one, the unstable scalarized stars always have the lowest binding energy in Fig. 2, even though there is no necessity for unstable equilibria to be energetically disfavored. This becomes a trivial point in Fig. 4 since a local maximum between two local minima has to have the highest energy. The list continues for other features like (i) the stable scalarized stars being metastable at lower $M_{\rm b}$ and becoming the global minimum only with increasing $M_{\rm b}$, (ii) the branch continuously connected to the GR solutions being unstable and (iii) the unstable scalarized stars always having lower scalar charge compared to the scalarized ones: $Q_- < Q_+$ (not plotted here).

The main factor controlling the order of the phase transition is the sign of the coefficient $b(M_{\rm b})$ in the Landau ansatz (3). For various theory parameters, we extracted the value of b at the point where the scalarized NSs branch off from the GR solutions, $b_0 = b(M_{\rm crit})$. See "End Matter" for details.

Fig. 5 clearly shows how b_0 becomes more negative as β becomes more negative for $m_\phi = 0$, the case being similar for all scalar masses. This is exactly in line with our observations in Figs. 3 and 4.

Discussion: We have demonstrated that first-order scalarization is the norm for our choice of STT action (1), but we also expect this to be a common occurrence. Scalarization is now known to be a generic phenomenon for various possible couplings between a scalar field and the metric [9, 12, 13, 42]. There are negative cases, as our preliminary analysis indicates that the coupling choice $A(\phi) = \cosh\left(\sqrt{3}\beta\phi\right)^{1/3\beta}$ [14, 43] features first-order scalarization less prominently. On the other hand, some other STTs feature first-order transitions even more prominently than Eq. (1). For example, scalarization of both black holes and NSs in scalar-Gauss-Bonnet theories are known to preferentially have first-order scalarization when couplings are explicitly enhanced in higher orders, e.g. $A(\phi) = e^{\beta\phi^2/2+\gamma\phi^4/4}$ [19–23].

Our phase transition picture already has a simple and powerful explanation for the above results. Changing the ϕ^4 term in the expansion of $A(\phi)$ directly modifies the higher order expansion coefficients in our Landau ansatz (3), specifically b. Recalling that the main difference between a first- and second-order transition is the sign of b, we can see how these higher order terms can determine the order of the phase transition. However, we emphasize that explicitly modifying the coupling or some form of *fine tuning* is not necessary for first-order scalarization by any means, as is clear from our choice of $A(\phi)$ here. Overall, we aim to study first-order scalarization for the most general STTs in order to devise more comprehensive tests for these theories.

In terms of observational prospects, the specific examples we gave are most relevant for lower mass NSs since

we concentrated on the onset of scalarization. There are recently proposed theoretical scenarios for these such as formation of NSs down to masses of $0.1M_\odot$ in accretion disks [44], and tidal disruption events in black hole–NS mergers where part of the NS mass can be ripped out [45, 46]. Such a change in the stellar mass can radically affect how large a scalar field can be supported by a star in general, and this becomes especially relevant for first-order scalarization where the discontinuous nature of the scalarization/descalarization process can provide distinct signals. There is already some work about first-order phase transitions near the maximum allowed NS masses, for which possible astrophysical signals have been studied [18]. For instance, a sudden transition between scalarized and non-scalarized configurations results in the emission of gravitational waves which carry two additional modes beyond those in GR. Moreover, these can also persist for longer periods. Together with related electromagnetic and neutrino signals from such a violent event, these gravitational waves offer distinct observational indicators for a first-order transition. Due to its discontinuous nature, first-order scalarization might also feature novel phenomenology in stellar collapse events, which have only been studied for the continuous second-order scalarization so far [11].

We reiterate that action (1) is the original and most popular scalarization theory in terms of the scalar coupling function $A(\phi)$, but it is still one specific choice. We expect other models of scalarization to have diverse characteristics, where different NS mass ranges are possible or even dominant for first-order scalarization. Studying these more general theories and how to test them via discontinuous phase transitions are major future research directions. Also note that the mass term m_ϕ makes almost all of our parameter space, the region with $m_\phi \gtrsim 10^{-15}$eV, unconstrained with current observations [9, 24]. We included the mostly-constrained $m_\phi = 0$ case in our analysis for the sake of completeness and also because it is still informative since the $m_\phi \to 0$ limit is continuous in terms of the NS structure.

The phase transition order depends on the sign of b. Let us remark that the parameters a, b, c, as well as the function M_0 and the critical mass $M_{\rm crit}$ depend on the parameters of the theory, (β, m_ϕ), and also the EOS. For simplicity let us focus on the β dependence, i.e. $b = b(M_{\rm b}, \beta)$. The fact that the order of the phase transition changes at some point as β becomes more and more negative means that b changes sign as β drops below a critical value $\beta_{\rm crit} < 0$, see Fig. 5. When $\beta = \beta_{\rm crit}$, we have a point where a and b vanish simultaneously, i.e., the point $(\beta = \beta_{\rm crit}, M_{\rm b} = M_{\rm crit}(\beta_{\rm crit}))$. Such a point is called a *tricritical point* [39, 40], and it separates a line of second-order phase transitions on the $(\beta, M_{\rm b})$ plane from a line of first-order ones. Spontaneous scalarization at $\beta_{\rm crit}$ features special characteristics whose study is another possible future research topic. Similarly, other aspects of the voluminous phase transition literature can be a source of other surprising results for gravitational phase transitions.

Lastly, we note that even though a phenomenological fit with very few parameters is possible, there is no understanding for the behavior of b_0 in Fig. 5 from first principles to the best of our knowledge. Recall that the scalar field profiles depend on the nonlinear terms that suppress the tachyonic instability, hence there is no simple method to relate the energy of the spacetime and the related Landau ansatz parameters to the STT parameters. Qualitatively, first-order scalarization becomes dominant when scalarization itself is stronger in terms of how much deviation we have from GR, which happens at higher $|\beta|$ [9, 24]. Whether this can lead us to the underlying reason for the change of sign in b_0 remains to be seen.

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